

Stellar PSFs Inflate Little Red Dots

Broad-band host measurements of compact high-redshift nuclei are systematically biased by mismatched stellar point-spread functions.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for compact nuclei.

First author / year / source

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Background

Compact high-redshift sources overlap with distinct physical components in diffraction-limited images. Field stars are commonly used to model the point-spread function and infer intrinsic structure.

This method assumes the target's spectrum matches the field stars, which is rarely true for little red dots.

What this paper is selling

Using little red dots as a test case, the authors show that broad-filter PSF fitting misassigns light when spectra differ. They derive the error's physical origin and demonstrate a correction for other compact distant sources.

Key results to remember

- A pure point source can appear to have a host galaxy.
- The spurious F444W host allocation exceeds the published 11.8% value.
- The bias changes inferred sizes, luminosities, and masses.

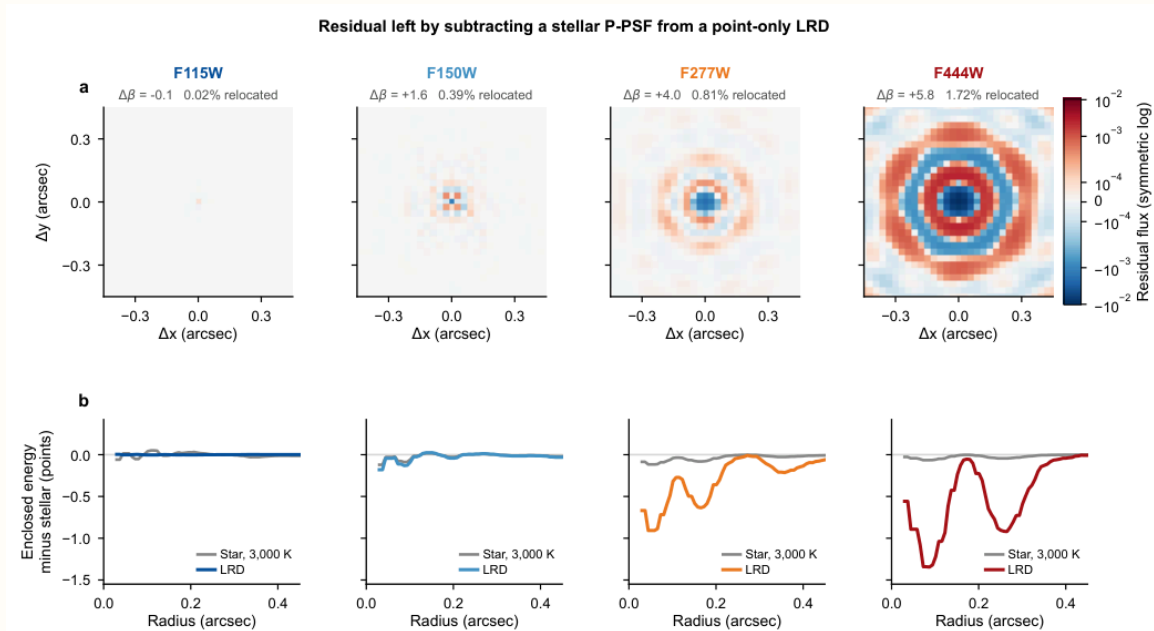


Fig. 1. A point-only little red dot produces wavelength-dependent residuals that mimic a Sérsic host component.

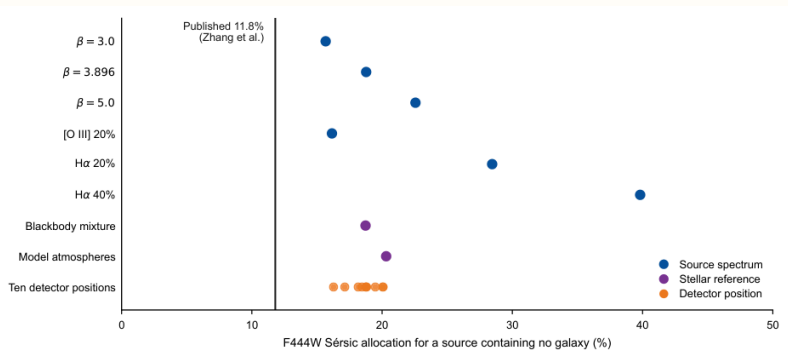


Fig. 2. Every tested input change leaves the spurious F444W host allocation above the published value.